

# TCP Connection Management

## (The Three-Way Handshake, Teardown, and State Transitions)

### 1. Connection Establishment: The Three-Way Handshake

Before segments can be sent, a TCP connection must be established. This process is crucial because it adds to perceived latency and can be a target for security attacks.

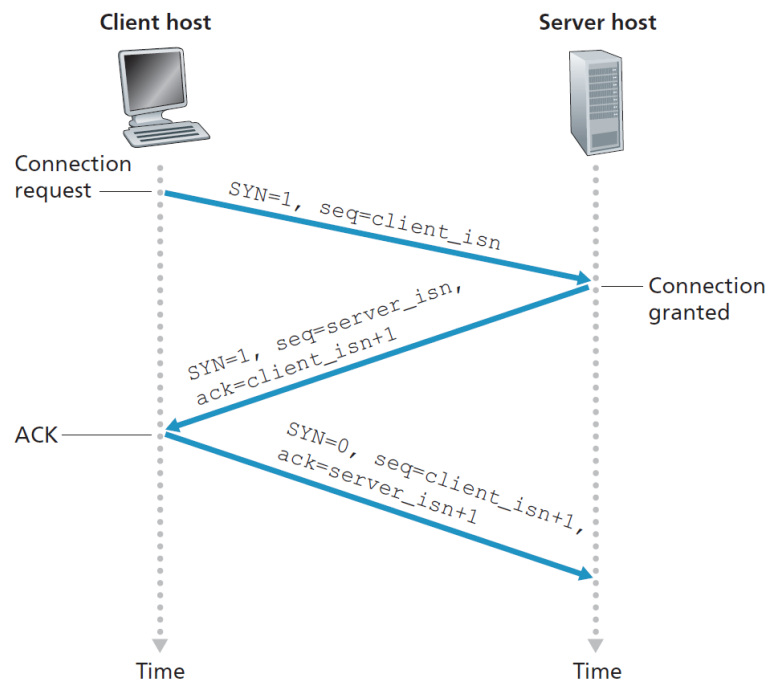


Figure 1: TCP Three-Way Handshake

#### The Three Steps

1. **Step 1: Client SYN** – The client sends a special segment with the **SYN bit = 1** and a randomly chosen initial sequence number (`client_isn`). No data is carried.
2. **Step 2: Server SYNACK** – The server extracts the SYN, allocates buffers/variables, and responds with a **SYNACK** segment (**SYN bit = 1**, **ACK field = `client_isn + 1`**). The server also chooses its own initial sequence number (`server_isn`).
3. **Step 3: Client ACK** – The client receives the SYNACK, allocates resources, and sends an ACK for the server's SYN (**ACK field = `server_isn + 1`**, **SYN bit = 0**). This segment may carry application data.

### Security: SYN Flood Attack

Attackers exploit Step 2 by sending many SYNs but never completing Step 3. The server's resources (buffers/variables) become exhausted waiting for ACKs that never arrive.

## 2. Connection Teardown

Either host can terminate the connection. When closed, all resources (buffers/variables) in both hosts are deallocated.

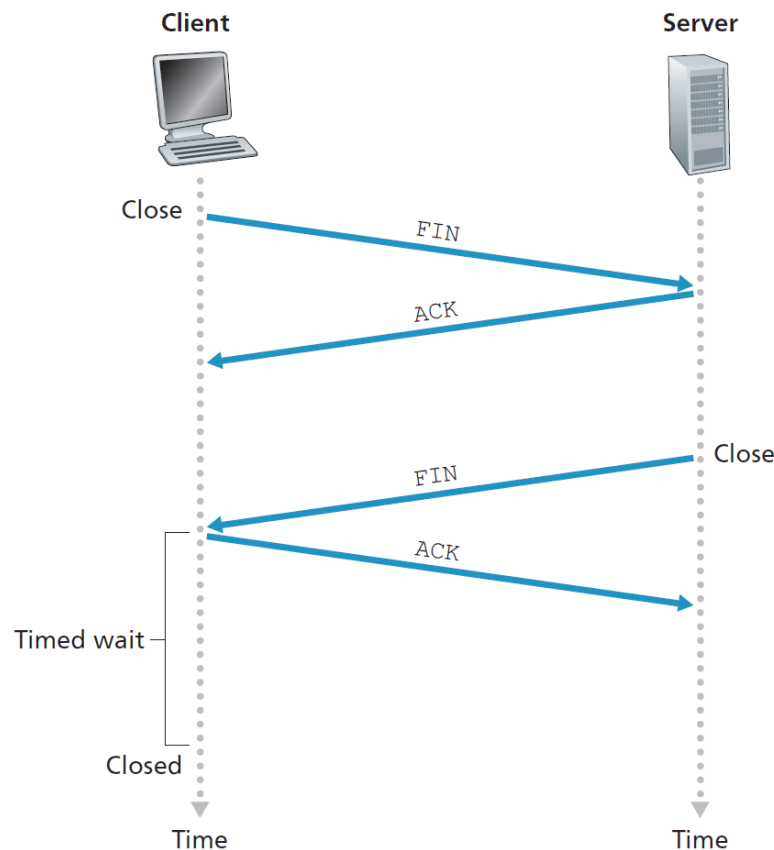


Figure 2: Closing a TCP Connection

### The Four-Step Teardown

1. **Initiate Close:** One side (e.g., Client) sends a segment with the **FIN bit = 1**.
2. **Server Acknowledgement:** The server responds with an **ACK** for the FIN.
3. **Server Shutdown:** The server then sends its own FIN segment.
4. **Final ACK:** The client responds with a final **ACK**, entering a **TIME\_WAIT** state before fully closing.

## 3. TCP State Transitions

During its lifecycle, a TCP connection visits several states.

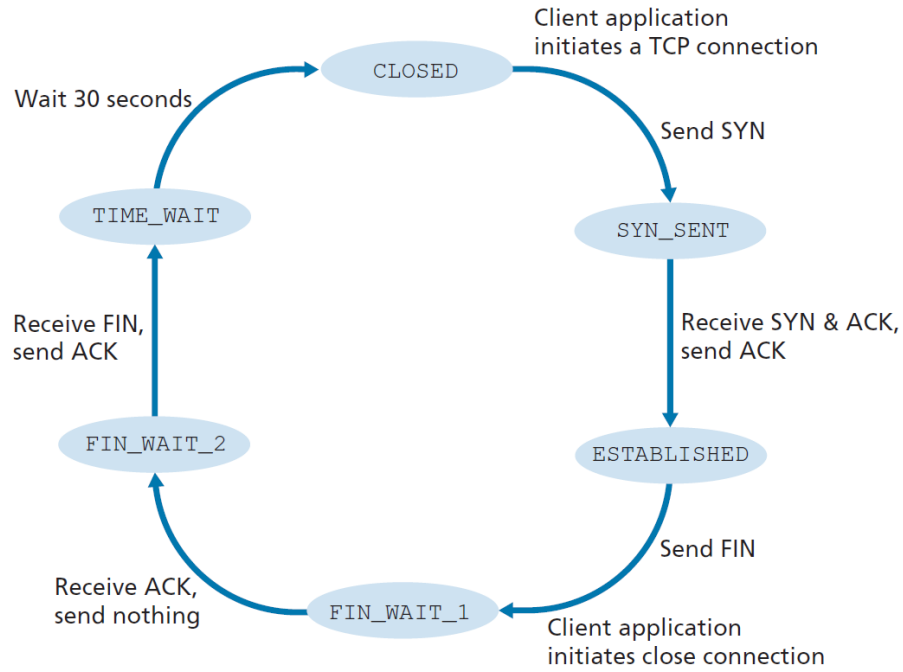


Figure 3: Client TCP State Transitions

### Key States

- **LISTEN**: Server is waiting for connection requests.
- **SYN\_SENT** / **SYN\_RCVD**: Handshake in progress.
- **ESTABLISHED**: Connection is active; data transfer is possible.
- **FIN\_WAIT\_1** / **FIN\_WAIT\_2**: Waiting for teardown acknowledgments.
- **TIME\_WAIT**: Ensures the final ACK is received by the other side (prevents "ghost" segments).

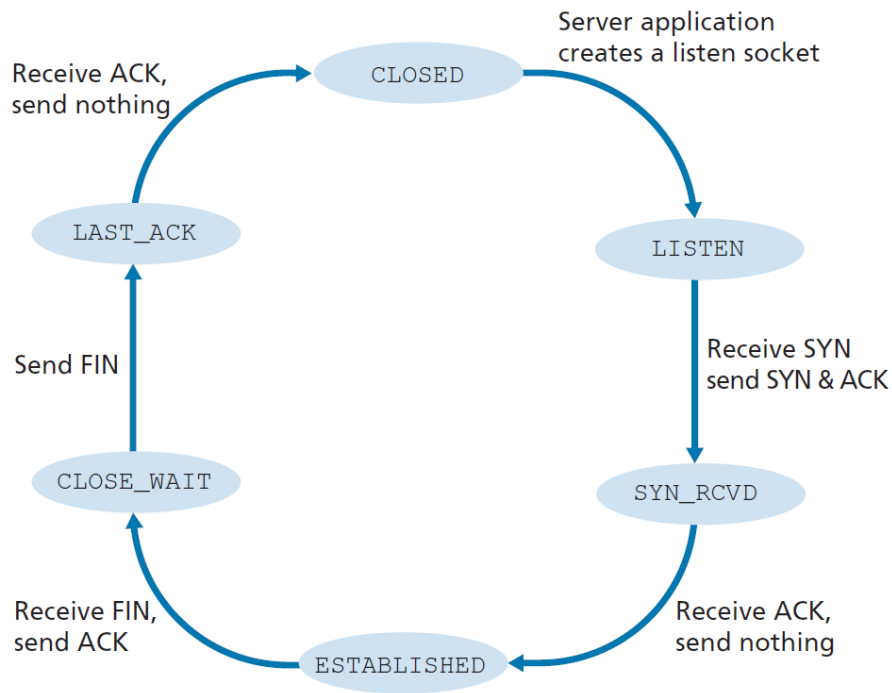


Figure 4: Server TCP State Transitions

## 4. Special Cases: RST Segments

If a host receives a segment whose port numbers do not match any ongoing socket (e.g., port 80 is closed):

- **TCP**: Sends a **reset segment** with the **RST bit = 1**.
- **UDP**: Sends a special **ICMP datagram**.

### Tool Spotlight: nmap

The **nmap** tool uses SYN, RST, and SYNACK segments to scan ports:

- **SYNACK received** → Port is **\*\*Open\*\***.
- **RST received** → Port is **\*\*Closed\*\*** (but host is reachable).
- **Nothing received** → Port is likely **\*\*Filtered\*\*** (firewall blocked the SYN).

## 5. Summary

### Connection Management Highlights

- Three-way handshake for robust establishment.
- Four-way sequence for connection teardown.
- Complex state machine ensures reliable sync and cleanup.
- RST segments handle pathological port-mismatch cases.